Tidal Asymmetry in Shallow Estuarine Basins

The measurements reported here were collected over nineteen consecutive spring tides at four stations along the northern shore. Each station carried a pressure sensor mounted on a steel frame, sampling at two-second intervals, and a paired turbidity probe fixed one metre above the bed. Instrument clocks were disciplined nightly against a shore reference, which bounds the timing uncertainty of any single record to under forty milliseconds.

Depth series were detided using a harmonic fit over eleven constituents. The residual, which we call the subtidal component, retains the storm surge signal and the seasonal steric adjustment. Because the basin is shallow relative to its length, friction dominates the momentum balance, and the flood limb of the curve steepens as the wave propagates inland. That steepening is the asymmetry this paper attempts to quantify, and it is visible without any filtering at the two inner stations.

Bed shear stress was not measured directly. We estimate it from the near-bed velocity profile under a logarithmic layer assumption, which holds for roughly eighty percent of the record and fails during the brief slack water intervals. Those intervals are excluded rather than interpolated, and the exclusion is marked in every figure so a reader can see where the estimate stops.

A referee asked whether the frame itself perturbs the flow. Scour marks around three of the four frames suggest that it does, at least locally. We repeated the innermost deployment with a smaller frame for a single fortnight and found the asymmetry index unchanged to within the scatter of that shorter record, which is weak evidence but the only evidence available.

Turbidity responded to the flood limb far more sharply than to the ebb. At the innermost station the suspended load during a rising tide exceeded the falling load by a factor of three, and the ratio grew with the tidal range. We interpret this as evidence that resuspension is controlled by the peak bed shear stress rather than by the integrated energy of the cycle.

Two caveats bound the conclusion. First, the turbidity probes saturate above roughly nine hundred formazin units, and saturation occurred on six occasions, all during the largest spring tides. Second, the northern shore receives a freshwater discharge whose gauge failed for eleven days in March. Neither gap falls inside the windows used for the regression, but both restrict how far the result may be extended toward the head of the estuary.

The comparison with the older survey deserves care. That campaign sampled hourly rather than continuously, and an hourly sample cannot resolve the peak of a flood limb lasting under ninety minutes. Where the two datasets overlap, the older one underestimates that peak by between twelve and thirty percent, and the discrepancy is largest exactly where the asymmetry is largest.

What follows for management is modest. If resuspension tracks peak stress, then dredging schedules built around mean energy will misallocate effort toward the calmer half of the cycle. We do not propose a revised schedule here; the basin geometry varies too much between reaches for any single rule, and four stations cannot support one.